

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO2: Experiment search and game-solving techniques such as Greedy Search, A, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)*

Assessment Tool Weightage: 40%

QUESTIONS: 3

Duration: 1 Hour
Total Marks:30

Annexure A

Dry Run Evaluation

Code of Conduct:

*I Thamish K (192412004) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: *Thil*

*25 + 24 + 23 + 12 + 10
= 94
DP*

Question - 6 - Nature of Environment

AI chess Robot

A chess playing robot can be classified as follows:

Property	Classification	Justification
Observability	Fully observable	The complete chessboard position
Determinism	Deterministic	A legal chess move produces a predictable board state
Episodic / sequential	Sequential	Each move affects future moves & the final result
Static / Dynamic	Dynamic	The environment changes whenever the opponent makes a move
Discrete / continuous	Discrete	Chess has a finite number of positions, moves & states.
Agents	Multi-Agent	There are two decision making agents: the robot & the opponent.

Final answer: Chess environment = Fully observable, deterministic, sequential, Dynamic, Discrete, Multi-Agent

2. Question 7 - Greedy Best-First Search

Given:

$$* h(A) = 6$$

$$* h(B) = 2$$

$$* h(C) = 4$$

$$* h(D) = 1$$

$$* h(E) = 3$$

$$* h(H) = 0$$

Greedy Best First Search uses:

$$f(n) = h(n)$$

It always select the node with the smallest heuristic value.

Step 1: Start at S

From S:

$$* A \rightarrow h(A) = 6$$

$$* B \rightarrow h(B) = 2$$

So select B because $2 < 6$

Step 2: Expand B

B connects to E.

Open List:

$$* E(3)$$

$$* A(6)$$

Select E

Step 3: Expand E

E becomes G.

Open heuristics:

* G(0)

* A(6)

select G.

G is the goal $\rightarrow$ Stop

Search Path:

$S \rightarrow B \rightarrow E \rightarrow G$

Priority Queue table

Iteration	Priority Queue	Expanded	$h(n)$
1	B(2), A(6)	S	-
2	E(3), A(6)	B	2
3	G(0), A(6)	E	3
4	-	G(goal)	0

3. Questions 8 - A* Search

A* uses:

$$F(n) = g(n) + h(n)$$

Given:

Node	g/h information
S	$h(S) = 0$
A	$h(A) = 2.5$
B	$h(B) = 3.4$
C	$h(C) = 4.3$
D	$h(D) = 5.2$
E	$h(E) = 6.1$
G	$h(G) = 8.0$

Assume each edge has cost 1

Step 1 - Expand S

$$g(S) = 0$$

For A:

$$g(A) = 1$$

$$f(A) = 1 + 2.5 = 3.5$$

For B:

$$g(B) = 1$$

$$f(B) = 1 + 3.4 = 4.4$$

OPEN:

A(3.5), B(4.4)

Choose A.

Step 2 - Expand A

A connects to C & D.

For C:

$$g(C) = 1$$

$$f(C) = 1 + 4 \cdot 3 = 6.3$$

For D:

$$g(D) = 1$$

$$f(D) = 1 + 5 \cdot 2 = 7.2$$

OPEN:

B(4.4), C(6.3), D(7.2)

Choose B.

Step 3 - Expand B

B connects to E.


$$g(E) = 2$$

$$f(E) = 2 + 6 \cdot 1 = 8.1$$

OPEN:

C(6.3), D(7.2), E(8.1)

Choose C.

Step 4 - Expand C

C connects to H.

$$g(H) = 3$$

$$f(H) = 3 + 8 \cdot 0 = 11.0$$

OPEN:

D(7.2), E(8.1), G(11.0)

Choose D.

Step 5. Expand D

D also connects to G.

Via D:

$$g(G) = 3$$

$$f(G) = 3 + 8 = 11$$

Same cost as already found.

OPEN:

E(8.1), G(11.0)

Choose E.

Step 6 - Expand E

E also connects to G.


$$g(G) = 3$$
$$f(G) = 3 + 8 = 11$$

OPEN:

G(11.0)

Choose G $\rightarrow$ Goal reached.

A* OPEN TABLE

The complete table is:

Iteration	Open	Expanded	g	h	f
1	A(3.5), B(4.4)	S	0	0	0
2	B(4.4), C(6.3), D(7.2)	A	1	2.5	3.5
3	C(6.3), D(7.2), E(8.1)	B	1	3.4	4.4
4	D(7.2), E(8.1), G(11.0)	C	2	4.3	6.3
5	E(8.1), G(11.0)	D	2	5.2	7.2
6	G(11.0)	E	2	6.1	8.1
7	-	G(goal)	3	8.0	11.0

A* search order:

